

Dementia manuscript | Mentor review

Prepared for Dr. Cockroft - September 7, 2026 - Internal draft, not sent

Dear Dr. Cockroft, please review the dementia component of our leakage-audited public brain MRI benchmark. The tumor specialist and single-head comparator remain part of the original study design.

Evidence retained

The accepted dementia specialist reached **0.9991 accuracy on 8,806 internal test images**; the conservative dHash sensitivity run reached 0.9975 on 8,881 images. The accepted single 8-class model remains slightly stronger than the hierarchical model (0.9972 vs 0.9963). These are public-image benchmark results, not clinical dementia diagnosis.

New explainability result: a negative control comparison

All 128 sampled images per specialist yielded valid Grad-CAMs. Masking the highest-CAM 20% of pixels reduced confidence **less** than equal-count random masking:

Specialist	Top-CAM drop	Random drop	Paired difference
Dementia	0.4453	0.6901	-0.2448
Tumor	0.3027	0.6808	-0.3780

This test does not support preferential sensitivity to the highlighted regions. Spatially different masks and out-of-distribution perturbations limit interpretation. The separate gallery contains seven dementia and eight tumor low-confidence/error examples, chosen by a fixed rule before inspecting maps. No expert anatomical validation was performed.

Boundaries unchanged

The dementia data are publicly sourced and augmented. Patient identifiers and acquisition metadata are absent; patient-level leakage and source shortcuts remain possible. Confidence is not calibrated clinical probability (integrated ECE: 0.1327 single-head; 0.1526 hierarchical). **No independent external cohort has been evaluated.** Accepted checkpoints were recovered and checked against archived predictions; no retraining or test-based tuning was performed.

Decisions requested

1. Keep the negative perturbation comparison and galleries as a transparent supplement?
2. Retain the combined benchmark framing, or emphasize dementia within that design?
3. Which venue and independent patient-level cohort should guide the next study?

Companion reading

Full manuscript: Methods 2.9, Results 3.6 and limitations. The companion bundle contains the browser gallery, Figures 8a-8b, quantitative summary, per-image evidence, provenance hashes and reproduction scripts. Start with **docs/review_bundle/README.md**. Authorship, venue formatting, formal references, image-use terms and institutional ethics wording still need completion.